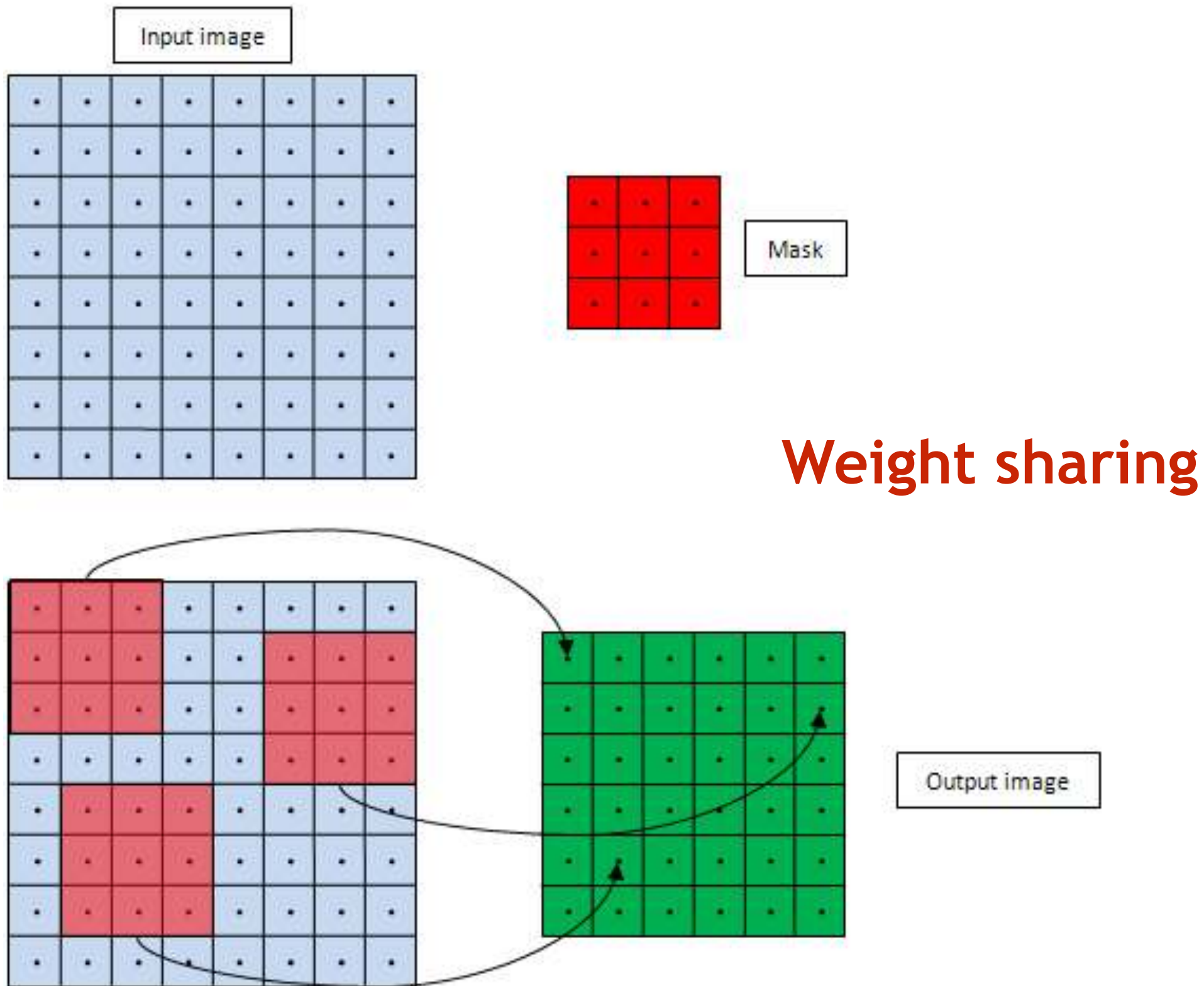


E9 205 Machine Learning for Signal Processing

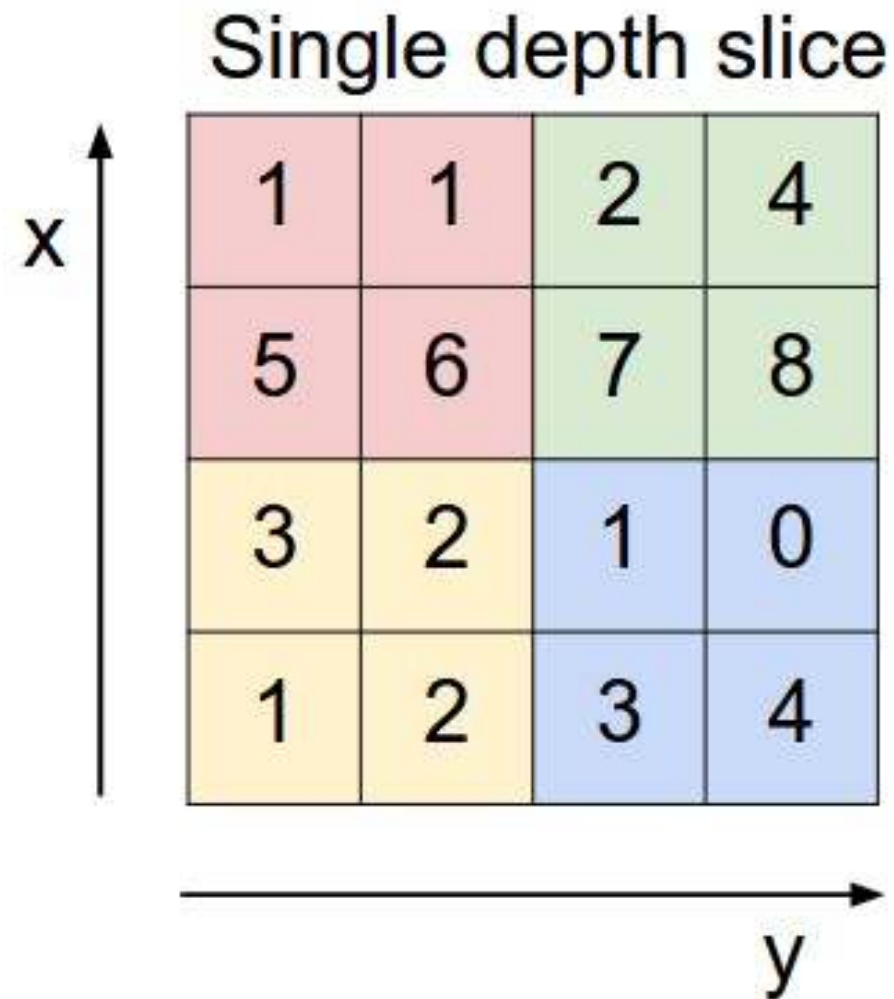
Convolutional Neural Networks

30-10-2019

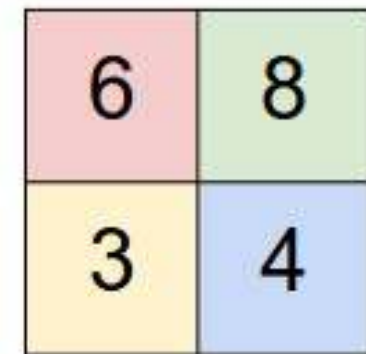
Other Architectures - Convolution Operation



Max Pooling Operation



max pool with 2x2 filters
and stride 2



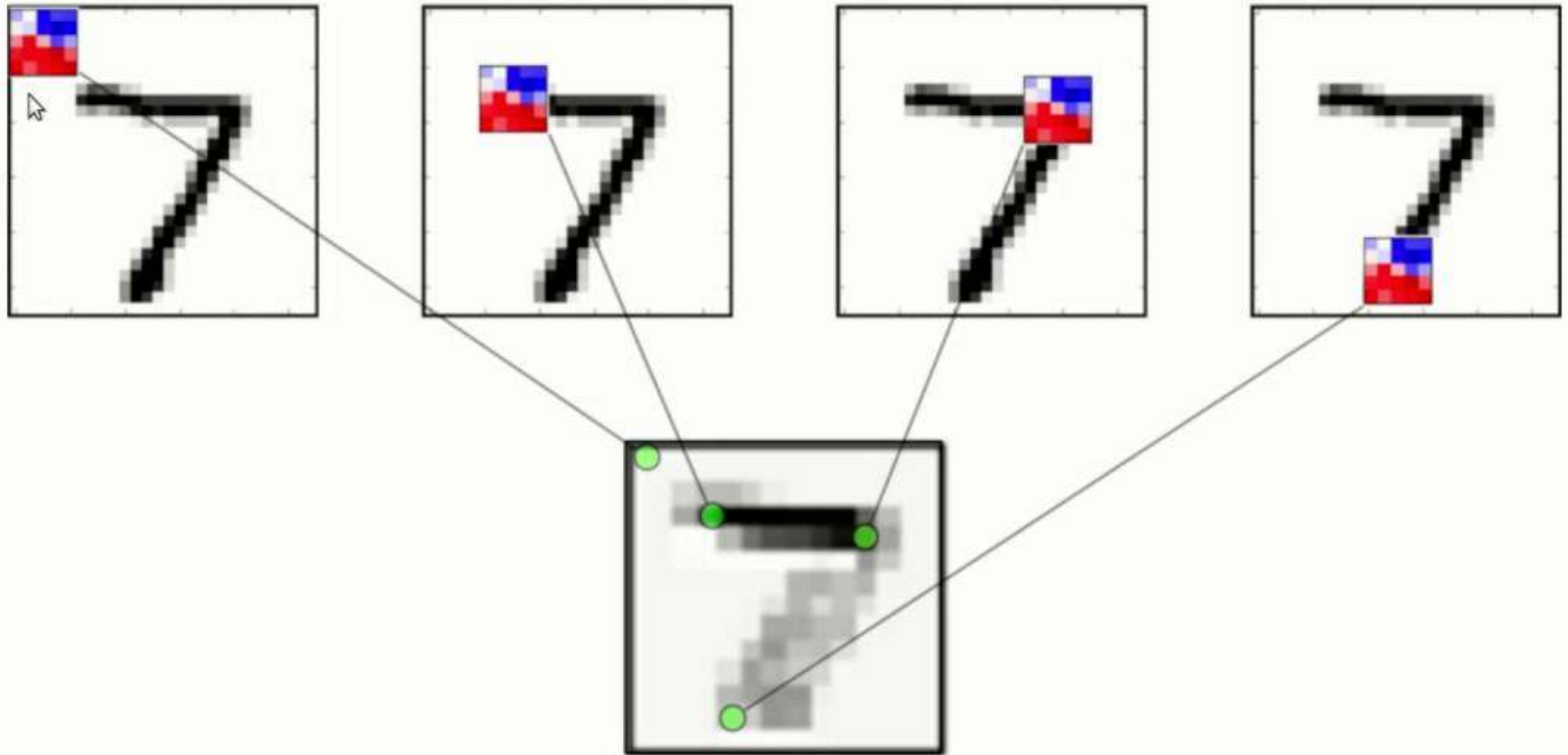
Convolution Operation (Images)

File Edit View Insert Cell Kernel Help Python 3

```
Image('images/02_convolution.png')
```

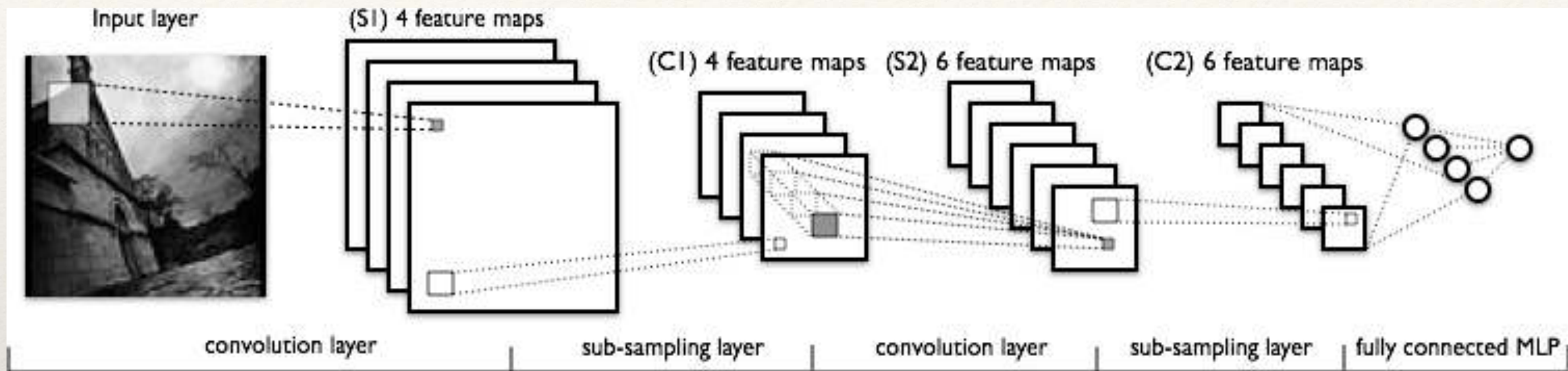
Out[2]:

Input Image with Filter Overlaid (4 copies for clarity)



Result of Convolution

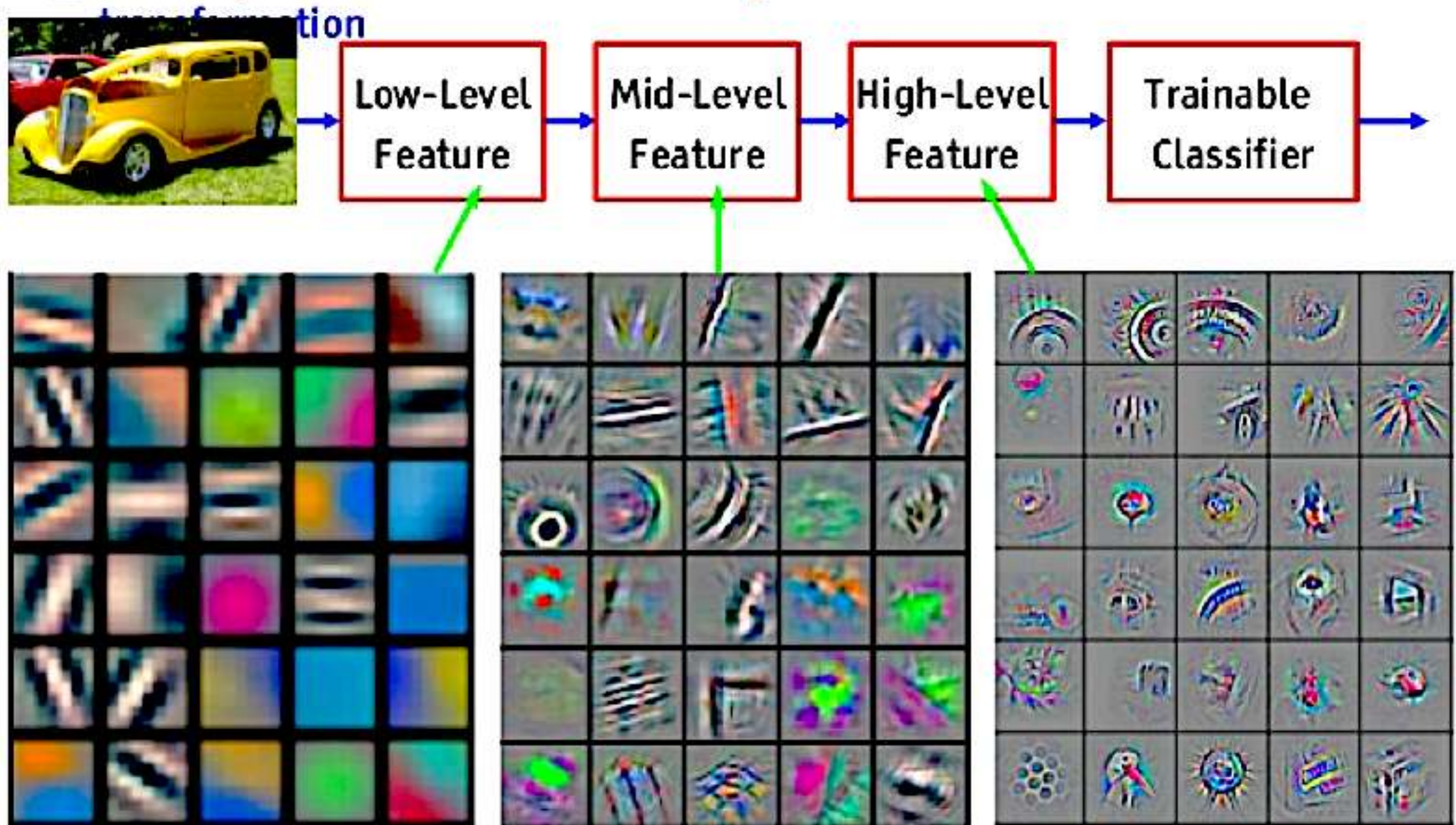
Convolutional Neural Networks



- Multiple levels of filtering and subsampling operations.
- Feature maps are generated at every layer.

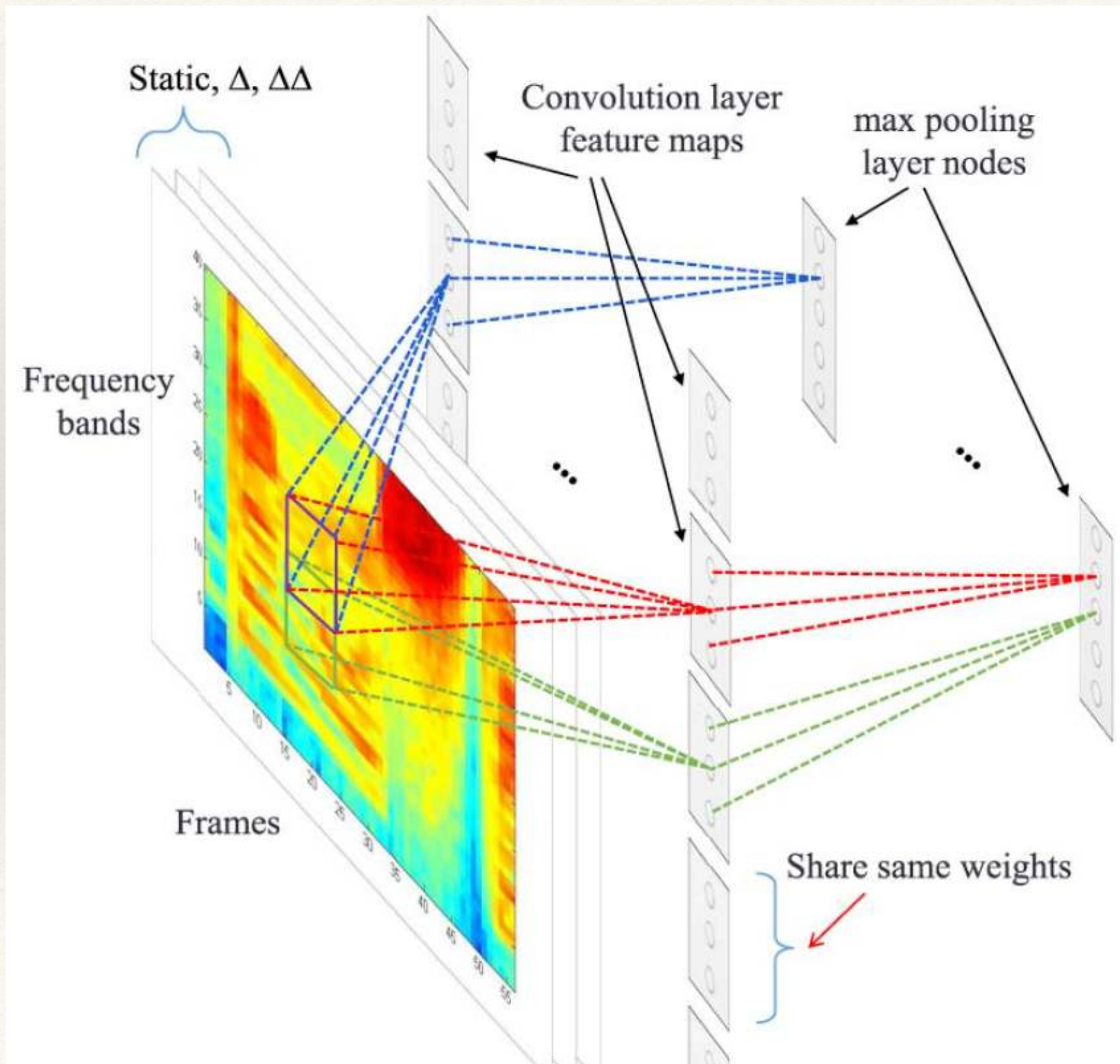
Representation Learning in CNNs

■ It's **deep** if it has **more than one stage** of **non-linear feature**



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

CNNs for Speech and Audio



Speech Recognition
Speaker/Language
Recognition

Convolutional Neural Networks on Images

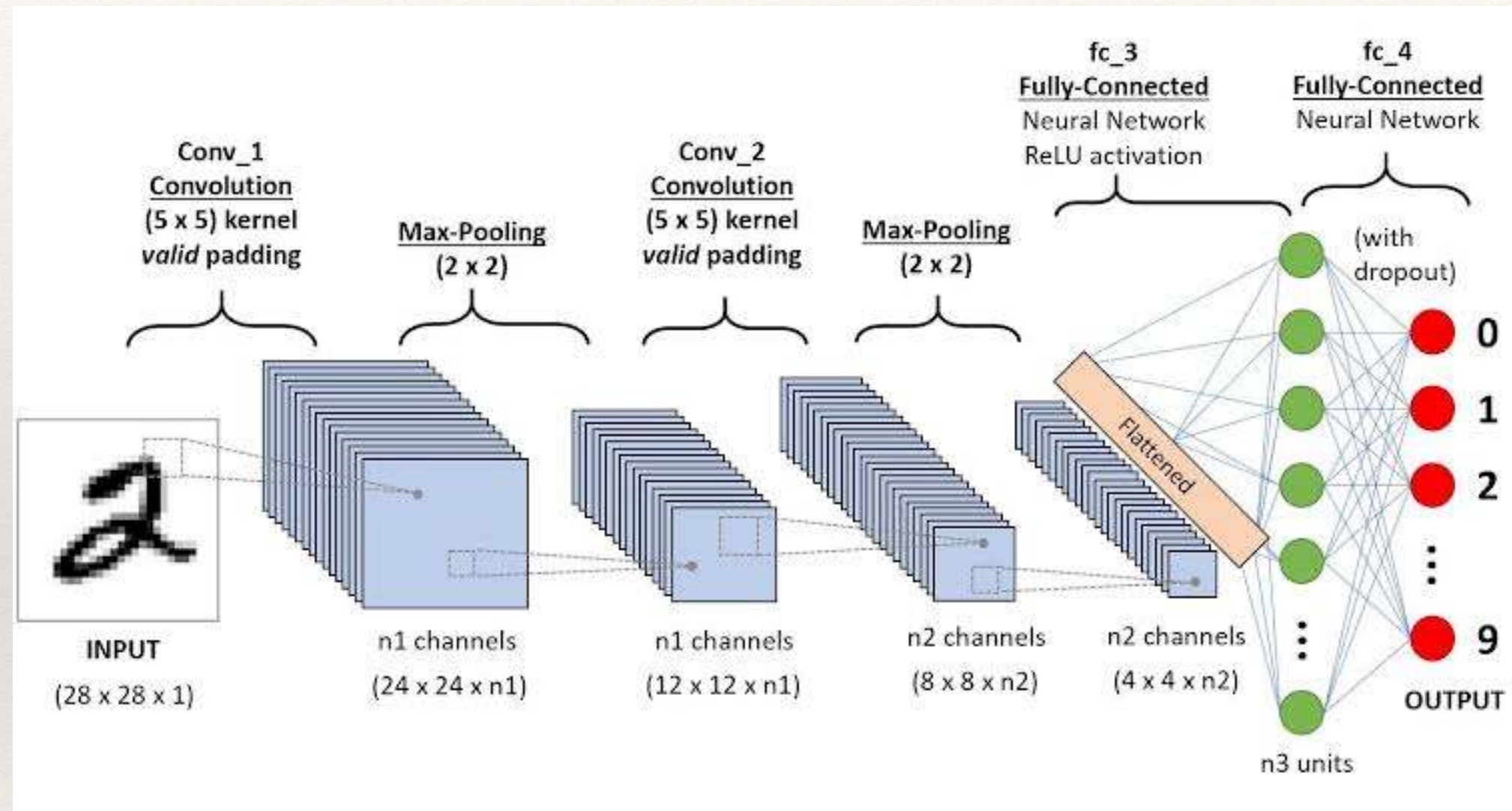


Image processing - Classification, segmentation, captioning, biomedical
Image processing applications.